

The minimal architecture of life: A rational reconstruction of biological structure

Abstract: What makes DNA structure the way that it is? Our paper provides a rational reconstruction of DNA structure, showing how the three central features of each DNA molecule are inferentially connected rather than merely co-discovered: the four-letter size of the DNA alphabet (T1), the aromatic six-membered ring structure of the DNA bases (T2), and the double-helical structure of the DNA polymer (T3). Beginning with a restricted set of empirical premises concerning strand antiparallelism, base pairing, aromaticity, X-ray diffraction geometry, and so on, we formulate a set of axioms that will allow us to derive T1–T3. The upshot of our argument is that the three central features of DNA structure are not independent empirical accidents but rather logically interdependent consequences of a minimal formal architecture.

Keywords: Rational reconstruction; DNA; antiparallel; Chargaff parity; Hückel's rule; aromatic; double helix

§1 Introduction

What makes DNA structure the way that it is? The standard textbook response would probably involve an appeal to the proposed double-helix structure made by Watson and Crick (1953), supported by the X-ray diffraction data of Franklin and Gosling (1953), and subsequent biochemistry confirming the four-base DNA alphabet, complementary base-pair geometry, and so on. According to this standard textbook view, the features of DNA structures are contingent empirical discoveries or things that happen to be true in a molecular-biological context, although they might, in principle, have been otherwise. Our paper argues that these appearances are misleading. Our central claim is that the principal features of DNA structure – its minimal alphabet size (§ 2), the aromaticity of its bases (§ 3), and its double-helical geometry (§ 4) – are not co-discovered facts but rather logically interdependent. Given a relatively small set of empirical and structural constraints, each follows from the others as a matter of formal necessity.

To make our claim explicit, we employ the method of rational reconstruction in the tradition of Carnap (1966), Hempel (1970), and Nagel (1979). More precisely, we characterize the structural theory of DNA as a finite set of theoretical postulates, from which the principal theorems of DNA structure T1–T3 (§§ 2–4) are derived. Our rational reconstruction aims therefore not at novelty of result, but rather at the transparency and clarity of inferential dependence.¹

¹ Following Carnap (1966), we distinguish between observational and theoretical vocabulary. Our empirical premises are stated in observational terms: measurements of distances in angstroms, spectrophotometric ratios, crystallographic patterns, and so on. Our axioms restate these empirical findings in theoretical terms and as formal structural constraints on any system that satisfies the relevant set of properties. Our theorems T1–T3 then follow as logical consequences of these axioms. In our rational reconstruction, empirical observation grounds abstraction, and abstraction grounds

§2 T1: The minimal DNA alphabet size is 4.

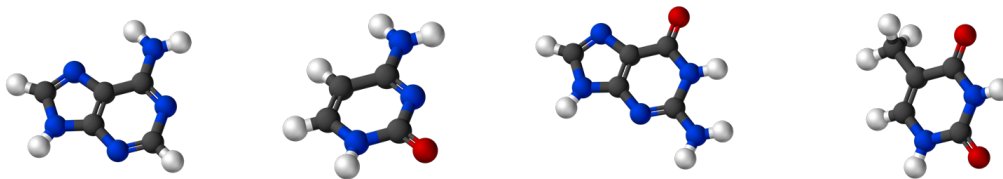


Figure 1: The four letters of the DNA alphabet (from left to right): adenine ($\text{C}_5\text{H}_5\text{N}_5$), cytosine ($\text{C}_4\text{H}_5\text{N}_3\text{O}$), guanine ($\text{C}_5\text{H}_5\text{N}_5\text{O}$), and thymine ($\text{C}_5\text{H}_6\text{N}_2\text{O}_2$). The ball-and-stick models have been coloured in accordance with the CPK (Corey-Pauling-Koltun) colouring scheme.

T1: The minimal DNA alphabet size is 4.

Empirical premises:

E1 (Uniform helix width): Each base pair spans exactly the same backbone-to-backbone distance of $\approx 10 \text{ \AA}$ in an ideal Watson-Crick model (Franklin and Gosling, 1953; Watson and Crick, 1953).

E2 (Two size classes): DNA bases fall into one of two structural classes: purines (bicyclic) and pyrimidines (monocyclic) (see also A2–A3 in § 3). In an ideal Watson-Crick model, a purine pairs with a pyrimidine to maintain the uniform helix width: the schematic widths are often given as $\approx 5.5 \text{ \AA}$ for purines and $\approx 3.5 \text{ \AA}$ for pyrimidines. A purine-purine pair is too wide for the backbone, while a pyrimidine-pyrimidine pair leaves a gap too large for hydrogen bonds ($\approx 2 \text{ \AA}$). Only purine-pyrimidine pairs satisfy the uniform helix width of E1.

E3 (Chargaff parity): For every species of double-stranded DNA,

formal deduction.

the DNA bases are present in equimolar quantities, as measured by UV spectrophotometry and paper chromatography. This holds across all organisms (Chargaff, 1950).

E4 (Antiparallel strand directionality) The two DNA strands run in opposite directions (see also A5 in § 4).

Premises or abstractions from E1–E4:²

P1 (Uniform helix width): X-ray crystallography of B-DNA shows that every base pair spans the same backbone-to-backbone distance of $\approx 10 \text{ \AA}$.

P2 (Two size classes): DNA bases fall into two discrete structural classes: purines (bicyclic and $\approx 5.5 \text{ \AA}$ in width) and pyrimidines (monocyclic and $\approx 3.5 \text{ \AA}$ in width). Only purine-pyrimidine pairs realize the uniform width of A1.

P3 (Chargaff parity): For every species of double-stranded DNA, the DNA bases are present in equimolar quantities. This holds across all organisms.

P4 (Antiparallel strand directionality) The two DNA strands run in opposite directions.

P5 (Sequence-encoded information) The sequence of DNA bases carries heritable biological information, not merely the length of the strand (Avery, MacLeod, and McCarty, 1944).³

Axioms or logical abstractions from P1–P5 into formal conditions:

A1 (Duplex structure): Each genetic polymer consists of two

² E1–E4 are physical facts, established by measurement, and they serve as the empirical foundation for our premises P1–P5.

³ The Avery–MacLeod–McCarty experiment showed that the transforming principle is the DNA rather than the proteins.

strands. Bases pair in a one-to-one manner across these two strands. – from P1 & P4

A2 (Size bipartition): The DNA alphabet $\mathcal{N}$ can be partitioned into two non-empty, disjoint subsets B (big bases) and S (small bases), such that $\mathcal{N} = B \cup S, B \cap S = \emptyset$. Valid base pairs are $(b, s) \in B \times S$: base pairing occurs only between a big base and a small base. – from P1 & P2

A3 (Specific complementarity): Each DNA base has a unique complement: there exists a function $f : \mathcal{N} \rightarrow \mathcal{N}$ such that $f(x)$ is the unique base that pairs with x . – from P3

A4 (Strand symmetry) Either strand can serve as a template. Formally, the function f is an involution $f \circ f = \text{id}$ that swaps size classes ($f(B) = S, f(S) = B$).⁴ – from P3 & P4

A5 (Non-trivial information): The alphabet $\mathcal{N}$ encodes information: – from P5

Lemmas:

L1: The function f is a fixed-point-free involution.

By A3, f is a function. By A2, every pair is in $B \times S$, so if $x \in B$, then $f(x) \in S$ and vice versa. Therefore, $f(x) \neq x$: no base pairs with itself. By A4, $f \circ f = \text{id}$. Therefore, f is a fixed-point-free involution. $\square$

L2: $|\mathcal{N}|$ is even.

A fixed-point-free involution on a finite set partitions it into disjoint two-cycles: $\{x, f(x)\}$. Therefore, $|\mathcal{N}| \in 2\mathbb{N}$. $\square$

⁴ An involution is a function that undoes itself when applied twice.

L3: $|B| = |S|$.

By A4, f maps B bijectively onto S . A bijection between finite sets implies equal cardinality: Set $p = |B| = |S|$. $\square$

L4: $p \geq 2$.

By A2 & A3, every element of B pairs with a unique element of S via f . Therefore, the valid base pairs are exactly p in number: $\{(b, f(b)) : b \in B\}$.⁵ By A5, we need at least two distinct base pairs to encode information, so $p \geq 2$. $\square$

Restatement of T1:

T1: Any duplex genetic polymer satisfying axioms A1–A5 has an alphabet $|\mathcal{N}| \geq 4$. Moreover, $|\mathcal{N}| = 4$ is achievable.

Proof of T1:

1. $|\mathcal{N}| = 2p$ – from L3
2. $p \geq 2$ – from L4
3. $\therefore |\mathcal{N}| \geq 4$ – from 1 & 2
4. $B = \{a, g\}, S = \{c, t\}$, with $f(a) = t, f(g) = c$ – assumption
5. This satisfies A1–A5: there are two strands, two size classes B and S , specific complementarity (the $a - t$ and $g - c$ base pairs), f is a fixed-point-free involution that allows size classes to be swapped, and two distinct pairs $\{(a, t), (c, g)\}$ allow for information to be encoded.
6. $\therefore |\mathcal{N}| = 4$ is achievable.

⁵ The number of distinct pair types equals p , since the function f is a bijection and A3 forces each large base b to have exactly one small partner s . By contrast, the total possible number of distinct valid pairs is $|B \times S| = p^2$.

Therefore, $|\mathcal{N}| \geq 4$ and $|\mathcal{N}| = 4$ is realized by DNA itself. $\square$

§3 T2: Every canonical DNA base contains at least one planar, aromatic, six-membered cycle

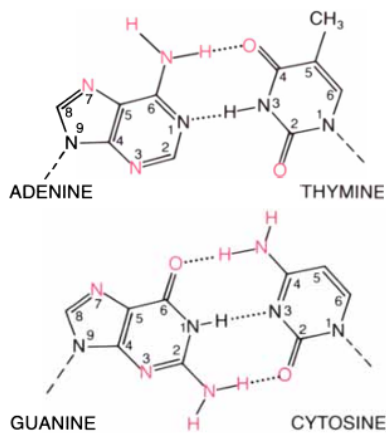


Figure 2: Base pairs shown in detail, with broken lines indicating bonds to sugar rings of the backbone. Adenine and guanine are purines (bicyclic), whereas cytosine and thymine are pyrimidines (monocyclic) (Dickerson, 1983, p. 97)

T2: Every canonical DNA base contains at least one planar, aromatic, six-membered cycle.

Empirical premises:

E1: DNA contains four canonical bases (see also T1 in § 2): adenine, cytosine, guanine, and thymine (Watson and Crick, 1953).

E2: Adenine and guanine are purine bases: bicyclic molecules with a six-membered ring fused to a five-membered ring (Labuda et al., 2018).

E3: Cytosine and thymine are pyrimidines: monocyclic molecules consisting of a single six-membered ring (Labuda et al., 2018).

E4: A ring is aromatic, just in case it is planar, cyclic, fully conjugated, and contains $4n + 2$ π -electrons for some non-negative integer $n \in \mathbb{Z}_{\geq 0}$ (Hückel, 1931a,b, 1932).⁶

E5: The six-membered rings in purines and pyrimidines satisfy Hückel’s rule with 6 (or $4 + 2$, where $n = 1$) π -electrons and are therefore planar and aromatic.

Axioms or logical abstractions from E1–E5:⁷

A1: $\forall x(x \in \{A, C, G, T\} \rightarrow \text{Base}(x))$ – definition of canonical DNA bases

A2: $\forall x(\text{Base}(x) \rightarrow \text{Purine}(x) \vee \text{Pyrimidine}(x))$ – partition into purines and pyrimidines

A3: $\neg \exists x (\text{Purine}(x) \wedge \text{Pyrimidine}(x))$ – the partition is exhaustive and disjoint

A4: $\forall x(\text{Purine}(x) \rightarrow \text{HasRing}(x))$ – purines contain a six-membered ring

A5: $\forall x(\text{Pyrimidine}(x) \rightarrow \text{HasRing}(x))$ – pyrimidines contain a six-membered ring

A6: $\forall x(\text{HasRing}(x) \rightarrow \exists r(\text{IsRing}(r) \wedge \text{PartOf}(r, x) \wedge \text{Planar}(r) \wedge \text{Aromatic}(r)))$ – every six-membered ring satisfies Hückel’s rule and is therefore planar and aromatic

⁶ π -electrons are electrons in covalent chemical bonds known as π or pi bonds. Examples of aromatic rings that obey Hückel’s rule include cyclopropenyl cations (2 π -electrons, where $n = 0$), benzene and DNA base rings (6 π -electrons, where $n = 1$), and so on.

⁷ Axioms A1–A6 abstract the empirical premises E1–E5 into the logical form required for the proof of T2. No further empirical assumptions are required.

Proof of T2:

1. $b \in \{A, C, G, T\}$ – assumption
2. $\text{Base}(b)$ – from $\forall_E$, A1, & 1
3. $\text{Purine}(b) \vee \text{Pyrimidine}(b)$ – from $\forall_E$, *modus ponens*, A2, & 2
4. $\text{Purine}(b)$ – assumption (case one)
5. $\text{HasRing}(b)$ – from $\forall_E$, *modus ponens*, A4, & 4
6. $\text{Pyrimidine}(b)$ – assumption (case two)
7. $\text{HasRing}(b)$ – from $\forall_E$, *modus ponens*, A5, & 6
8. $\text{HasRing}(b)$ – from $\forall_E$, A3, and 3–7
9. $\text{HasRing}(b) \rightarrow \exists r(\text{IsRing}(r) \wedge \text{PartOf}(r, b) \wedge \text{Planar}(r) \wedge \text{Aromatic}(r))$
– from $\forall_E$ & A6
10. $\exists r(\text{IsRing}(r) \wedge \text{PartOf}(r, b) \wedge \text{Planar}(r) \wedge \text{Aromatic}(r))$ – from
modus ponens & 8–9
11. $\text{HasRing}(b) \wedge \exists r(\text{IsRing}(r) \wedge \text{PartOf}(r, b) \wedge \text{Planar}(r) \wedge \text{Aromatic}(r))$
– from $\wedge_I$, 8, & 10
12. $\therefore \forall x(\text{Base}(x) \rightarrow \text{HasRing}(x) \wedge \exists r(\text{IsRing}(r) \wedge \text{PartOf}(r, x) \wedge$
 $\text{Planar}(r) \wedge \text{Aromatic}(r)))$ – from $\forall_I$, $\rightarrow_I$, & 1–11

Therefore, every canonical DNA base contains at least one planar, aromatic, six-membered ring. $\square$

§4 T3: The DNA molecule has a double-helical structure.

T3: The DNA molecule has a double-helical structure.

Assumptions or empirical premises:⁸

⁸ A1 is the only assumption requiring proof. We have already established the proof of A1 from first principles via Hückel's rule in § 3. A2–A5 are empirical premises, supported

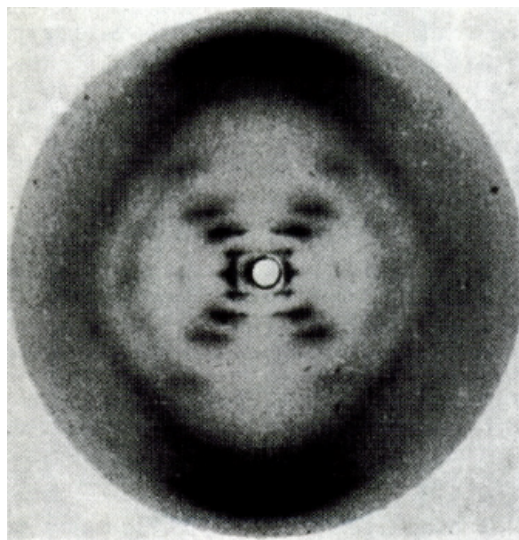


Figure 3: Photo 51 or an X-ray based fibre diffraction image of a paracrystalline gel composed of DNA fibre (Franklin and Gosling, 1953, p. 740)

A1 (Planarity): Each DNA base is a planar aromatic figure. – from T2 (proven in § 3)

A2 (Rise): Successive bases are separated by $d = 3.4 \text{ \AA}$ along the axis, where $\text{\AA}$ denotes an angstrom or unit of length equal to 10^{-10} m (Franklin and Gosling, 1953).

A3 (Twist): In the model of B-DNA (the standard double-helix form of DNA found under ordinary biological conditions), there are about 10 DNA bases per turn (Franklin and Gosling, 1953). Therefore, each base pair is rotated by $\theta = (\frac{360}{10})^\circ = 36^\circ$ from the previous base pair in B-DNA.

A4 (Radius): Each glycosidic attachment lies at $r \approx 10 \text{ \AA}$ from

by Franklin's (1953) X-ray diffraction data and Watson and Crick's (1953) base-pairing geometry. The double helix structure is therefore not a shape that DNA molecules happen to have: it is a natural geometric consequence arising from the combined constraints of base planarity (due in part to aromaticity), base stacking, backbone geometry, complementary pairing, and so on.

the axis (Franklin and Gosling, 1953).

A5 (Pairing): Two DNA strands are antiparallel, and the DNA base pairs form nearly perpendicular bridges between them (Watson and Crick, 1953).

Lemmas:

L1: A single DNA strand traces a right-handed helix.

Place the helical axis along the z -axis. A2–A4 then determine the n^{th} base attachment point uniquely as:

$$P_n = (r \cos(n\theta), r \sin(n\theta), nd)$$

Each time n increases by 1, the point rotates by an angle θ around the z -axis and moves up by a distance d along it. For $\theta > 0$ and $d > 0$, this will trace a right-handed helix.⁹ Therefore, P_n lies on the parametric equation of a right-handed helix. Furthermore and through A1, a planar aromatic disc is attached at each P_n perpendicular to the z -axis. $\square$

L2: Two antiparallel DNA strands form a double helix.

A second strand satisfying the same r , d , $|\theta|$ but opposite rotational sense has the n^{th} base attachment point as:

$$Q_n = (r \cos(-n\theta + \varphi), r \sin(-n\theta + \varphi), nd),$$

⁹ The degenerate cases are $\theta = 0$ and $d = 0$. When $\theta = 0$, there is no rotation from one step to the next, so all points $P_n = (r, 0, nd)$ lie on the same vertical line, yielding a straight line rather than a helix. When $d = 0$, all points $P_n = (r \cos(n\theta), r \sin(n\theta), 0)$ stay on the same plane, yielding a circle or planar orbit rather than a helix.

where $\varphi \approx 180^\circ$ is fixed by the hydrogen-bond bridge length, which forces the two attachment points P_n and Q_n to sit on opposite sides of the axis (A5). Each Q_n sits at the same height nd as P_n , connected by a perpendicular base-pair rung (A5). Two coaxial helices of opposite sense, linked by perpendicular base-pair rungs, constitute a double helix. $\square$

Proof of T3:

1. Each canonical DNA base is planar and aromatic. – from A1 or T2 (proof discharged in § 3)
2. d , θ , and r are fixed by the backbone geometry. – from A2–A4
3. A single DNA strand traces a right-handed helix. – from L1 & 1–2
4. Two DNA strands run antiparallel with perpendicular base-pair rungs. – from A5
5. The second DNA strand traces a coaxial helix. – from L2 & 2–4
6. $\therefore$ The two DNA strands together form a double helix. – from $\wedge_I$, 3, & 5

Therefore, DNA is a double helix. $\square$

§5 Conclusion

In conclusion, we have employed the method of rational reconstruction to make sense of DNA structure. Our reconstruction is local and targeted rather than grand and sweeping. Its focus is on three theorems about DNA structure:

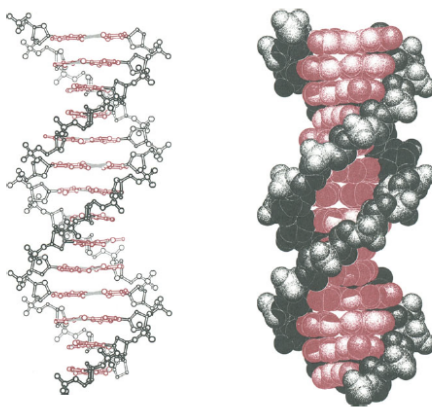


Figure 4: B-DNA structure (Saenger, 1984, p. 262)

- T1: The minimal DNA alphabet size is 4 (§ 2);
- T2: Every canonical DNA base contains at least one planar, aromatic, six-membered cycle (§ 3);
- T3: The DNA molecule has a double-helical structure (§ 4).

Our paper has been concerned with what the minimal formal apparatus might look like that will allow us to derive each of these theorems T1–T3 and what logical relationships might obtain between T1–T3. Our results are illuminating. In § 2, we have shown that any duplex genetic polymer satisfying five structural axioms A1–A5 (duplex structure, size partition, specific complementarity, strand symmetry, and non-trivial information) must have an alphabet of at least four bases, with four being achievable by the DNA alphabet itself. In § 3, we have shown that the aromaticity of the four canonical DNA bases (as established in § 2) follows from Hückel’s rule and the classification of these bases as either purines or pyrimidines. In § 4, we have shown that the double-helical geometry of DNA follows from base planarity (as established in § 3) and the empirical parameters of rise, twist, radius, and pairing as determined by X-ray crystallography. Theorems T1–T3 are

therefore not logically independent: the double helix structure of DNA is logically downstream of Chargaff parity (T1) and Hückel’s rule (T2).

The inferential nestedness is the key contribution of our paper: T3 depends on T2, and both T2 and T3 depend on the structural constraints associated with T1. Our rational reconstruction does not overturn any biological fact. Instead, it reveals that empirical facts about DNA structure are more tightly connected than their history of empirical discovery might suggest. The logical space of possible genetic polymers satisfying the minimal structural constraints identified in our paper, for instance, is far more constrained than a contingentist reading of molecular biology might imply.

References

- Avery, Oswald T., Colin M. MacLeod, and Maclyn McCarty (1944). “Studies on the Chemical Nature of the Substance Inducing Transformation of Pneumococcal Types: Induction of Transformation by a Deoxyribonucleic Acid Fraction Isolated from *Pneumococcus* Type III”. In: *Journal of Experimental Medicine* 79.2, pp. 137–158.
- Carnap, Rudolf (1966). *Philosophical Foundations of Physics*. Vol. 966. Basic Books.
- Chargaff, Erwin (1950). “Chemical specificity of nucleic acids & mechanism of their enzymatic degradation”. In: *Experientia* 6.6, pp. 201–209.
- Dickerson, Richard E. (1983). “The DNA helix & how it is read”. In: *Scientific American* 249.6, pp. 94–111.
- Franklin, Rosalind E. and Raymond G. Gosling (1953). “Molecular configuration in sodium thymonucleate”. In: *Nature* 171.4356, pp. 740–741.
- Hempel, Carl G. (1970). *Aspects of Scientific Explanation*. Vol. 965. Free Press.

- Hückel, Erich (1931a). “Quantentheoretische Beiträge zum Benzolproblem II. Quantentheorie der induzierten Polaritäten”. In: *Zeitschrift für Physik* 72.5-6, pp. 310–337. DOI: 10.1007/BF01341953.
- (1931b). “Quantentheoretische Beiträge zum Benzolproblem I. Die Elektronenkonfiguration des Benzols und verwandter Verbindungen”. In: *Zeitschrift für Physik* 70.3-4, pp. 204–286. DOI: 10.1007/BF01339530.
- (1932). “Quantentheoretische Beiträge zum Problem der aromatischen und ungesättigten Verbindungen. III”. In: *Zeitschrift für Physik* 76.9-10, pp. 628–648. DOI: 10.1007/BF01341936.
- Labuda, Ján et al. (2018). In: *Pure & Applied Chemistry* 90.7, pp. 1121–1198. DOI: doi:10.1515/pac-2016-1120. URL: <https://doi.org/10.1515/pac-2016-1120>.
- Nagel, Ernest (1979). *The Structure of Science*. Vol. 411. Hackett Publishing Company Indianapolis.
- Saenger, Wolfram (1984). *Principles of Nucleic Acid Structure*. Springer.
- Watson, James D. and Francis Crick (1953). “Molecular structure of nucleic acids: A structure for deoxyribose nucleic acid”. In: *Nature* 171.4356, pp. 737–738.